

Beyond the One-Hour Standard

Research Presentation

Using **Digital Trace Data**
to Reconstruct Nigeria's
Discontinued
Unemployment Series

The Problem: Numerical vs. Actual Reality



The 2023 shift to a **1-hour employment standard** created a "numerical reality" that masks true underemployment and informal labor distress in Nigeria.

Core Concepts

- Labor Utilization Framework
- Digital Trace Theory
- Job Search Theory

Core Concept

Utilizing "**unconscious disclosure**" from digital trace data as a revealed preference of actual job-seeking effort.

Research Question



Does **high-frequency digital trace data** possess the predictive power required to reliably reconstruct the discontinued **40-hour unemployment series** in Nigeria?

Methodological Framework



Objective: To empirically reconstruct the discontinued 40-hour unemployment baseline across the 2023 institutional rupture.

Stage 1: Feature Generation

LLM-driven selection of culturally contextualized keywords (excluding Western welfare terms).

Stage 2: Dim. Reduction

Elastic Net regularization to objectively filter noise and resolve severe multicollinearity.

Stage 3: Reconstruction

Regularized machine learning (Ridge Regression/XGBoost) combined with temporal splicing.

Data & Evaluation Strategy

Target Data (The Anchor)

Historical household survey data (2004–2020) from the Nigerian National Bureau of Statistics (NBS), aligned via temporal disaggregation.

Predictor Data (The Signal)

High-frequency Google Trends search volumes capturing formal job seeking, survival entrepreneurship, and emigration intent.

Evaluation Methodology



Walk-forward validation and Diebold-Mariano testing to prove statistically significant predictive superiority over baseline models.